

## HW #10

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**Question 1.** In Exercise 4.6, we have determined all 7 quadratic extensions of  $\mathbb{Q}_2$ . For each quadratic extension  $K/\mathbb{Q}_2$ , compute the local conductor  $f_{K/\mathbb{Q}_2}$ .

*Proof.* We found that  $\mathbb{Q}_2(\sqrt{\pm 2})$ ,  $\mathbb{Q}_2(\sqrt{\pm 6})$ ,  $\mathbb{Q}_2(\sqrt{\pm 3})$ ,  $\mathbb{Q}_2(\sqrt{-1})$  are the 7 quadratic extensions of  $\mathbb{Q}_2$ . Note that, for a quadratic extension  $K/\mathbb{Q}_2$ ,  $N_{K/\mathbb{Q}_2}(K^\times) \supset (\mathbb{Q}_2^\times)^2$ . By Exercise 4.6, we know that  $(\mathbb{Q}_2^\times)^2 \supset 1 + 8\mathbb{Z}_2$ . Therefore,  $f_{K/\mathbb{Q}_2} \leq 3$ . Note also that if  $K/\mathbb{Q}_2$  is ramified, then  $N_{K/\mathbb{Q}_2}(\mathcal{O}_K^\times)$  is an index 2 subgroup of  $\mathbb{Z}_2^\times = 1 + 2\mathbb{Z}_2$ , so  $f_{K/\mathbb{Q}_2} \geq 2$ . Thus,  $f_{K/\mathbb{Q}_2}$  can only possibly be 0, 2, 3. To check whether  $f_{K/\mathbb{Q}_2}$  is 0, 2, or 3, we only need to check the following:

- if  $K = \mathbb{Q}_2(\sqrt{5})$ ,  $f_{K/\mathbb{Q}_2} = 0$ ;
- otherwise, if 5 (mod 8) is in the norm group,  $f_{K/\mathbb{Q}_2} = 2$ ; if not,  $f_{K/\mathbb{Q}_2} = 3$ .

Now we check.

- $K = \mathbb{Q}_2(\sqrt{-2})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 + 2y^2 : x, y \in \mathbb{Z}_2\}$$

Note that a square is  $\equiv 0, 1, 4 \pmod{8}$ , so  $x^2 + 2y^2 \equiv 0, 1, 2, 3, 4, 6 \pmod{8}$ , so it cannot be 5 (mod 8). Therefore,  $f_{K/\mathbb{Q}_2} = 3$ .

- $K = \mathbb{Q}_2(\sqrt{2})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 - 2y^2 : x, y \in \mathbb{Z}_2\}$$

Note that a square is  $\equiv 0, 1, 4 \pmod{8}$ , so  $x^2 - 2y^2 \equiv 0, 1, 2, 4, 6, 7 \pmod{8}$ , so it cannot be 5 (mod 8). Therefore,  $f_{K/\mathbb{Q}_2} = 3$ .

- $K = \mathbb{Q}_2(\sqrt{-6})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 + 6y^2 : x, y \in \mathbb{Z}_2\}$$

Note that a square is  $\equiv 0, 1, 4 \pmod{8}$ , so  $x^2 + 6y^2 \equiv 0, 1, 2, 4, 6, 7 \pmod{8}$ , so it cannot be 5 (mod 8). Therefore,  $f_{K/\mathbb{Q}_2} = 3$ .

- $K = \mathbb{Q}_2(\sqrt{6})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 - 6y^2 : x, y \in \mathbb{Z}_2\}$$

Note that a square is  $\equiv 0, 1, 4 \pmod{8}$ , so  $x^2 - 6y^2 \equiv 0, 1, 2, 3, 4, 6 \pmod{8}$ , so it cannot be 5 (mod 8). Therefore,  $f_{K/\mathbb{Q}_2} = 3$ .

- $K = \mathbb{Q}_2(\sqrt{-3}) = \mathbb{Q}_2(\sqrt{5})$  has  $f_{K/\mathbb{Q}_2} = 0$  as we already know  $K/\mathbb{Q}_2$  is unramified.

- $K = \mathbb{Q}_2(\sqrt{3})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 - 3y^2 : x, y \in \mathbb{Z}_2\}$$

Note that  $N_{K/\mathbb{Q}_2}(\sqrt{3}) = -3$ , so  $f_{K/\mathbb{Q}_2} = 2$ .

- $K = \mathbb{Q}_2(\sqrt{-1})$ :

$$N_{K/\mathbb{Q}_2}(\mathcal{O}_K) = \{x^2 + y^2 : x, y \in \mathbb{Z}_2\}$$

Note that  $N_{K/\mathbb{Q}_2}(2 + \sqrt{-1}) = 5$ , so  $f_{K/\mathbb{Q}_2} = 2$ .

□

**Question 2.** Use local class field theory to show the following.

- (1) Let  $L/K$  be a totally ramified abelian extension of local fields. Suppose that  $L/K$  is tamely ramified. Let  $k_K$  be the residue field of  $K$ . Show that  $[L : K] \leq \#k_K - 1$ . If  $[L : K] = \#k_K - 1$ , we say that  $L$  is a **maximal totally tamely ramified abelian extension** of  $K$ .
- (2) Show that there are exactly  $\#k_K - 1$  many distinct maximal totally tamely ramified abelian extensions of  $K$ .<sup>1</sup>
- (3) We know that  $\mathbb{Q}_3(\zeta_3)$  is a maximal totally tamely ramified abelian extension of  $\mathbb{Q}_3$ . Find another maximal totally ramified abelian extension of  $\mathbb{Q}_3$  that is not isomorphic to  $\mathbb{Q}_3(\zeta_3)$ .

**Hint.** You may find Exercise 4.6 helpful.

*Proof.*

- (1) As  $L/K$  is totally ramified,  $[L : K] = e_{L/K}$ . We know that  $e_{L/K} = [\mathcal{O}_K^\times : N_{L/K}(\mathcal{O}_L^\times)]$ . Let  $\pi_K$  be a uniformizer of  $K$ . Furthermore, as  $L/K$  is tamely ramified, we know that  $N_{L/K}(\mathcal{O}_L^\times) \supset 1 + \pi_K \mathcal{O}_K$ . Therefore,  $e_{L/K} \leq [\mathcal{O}_K^\times : 1 + \pi_K \mathcal{O}_K] = \#k_K^\times = \#k_K - 1$ .
- (2) By local class field theory, it is the same as counting the number of subgroups  $G$  of  $K^\times$  of index  $\#k_K - 1$  such that  $G \cap \mathcal{O}_K^\times = 1 + \pi_K \mathcal{O}_K$ . Note that the subgroup  $\pi_K^\mathbb{Z} \times (1 + \pi_K \mathcal{O}_K) \subset K^\times$  depends on the choice of  $\pi_K$ , so there are possibly many of them.

Let us choose a uniformizer  $\pi_K$ , and let  $N := \#k_K - 1$ . Note that, if  $G \leq K^\times$  is an index  $N$  subgroup, then  $G \supset 1 + \pi_K \mathcal{O}_K$ . Therefore, the number of index  $N$  subgroups of  $K^\times$  whose intersections with  $\mathcal{O}_K^\times$  being  $1 + \pi_K \mathcal{O}_K$  is the same as the number of index  $N$  subgroups of  $K^\times / (1 + \pi_K \mathcal{O}_K)$  which intersect trivially with  $\mathcal{O}_K^\times / (1 + \pi_K \mathcal{O}_K)$ . Note we can identify

$$K^\times / (1 + \pi_K \mathcal{O}_K) = \pi_K^\mathbb{Z} \times K^\times / (1 + \pi_K \mathcal{O}_K) \cong \mathbb{Z} \times \mathbb{Z} / N\mathbb{Z}$$

as abelian groups. Therefore, this is also the same as counting the number of index  $N$  subgroups of  $\mathbb{Z} \times \mathbb{Z}$  which intersect with  $0 \times \mathbb{Z}$  into  $0 \times N\mathbb{Z}$ . This perspective is nice as we know that any finite index subgroup of  $\mathbb{Z} \times \mathbb{Z}$  is isomorphic to  $\mathbb{Z} \times \mathbb{Z}$ , so it is about finding two vectors  $v, w \in \mathbb{Z}^2$  that span an index  $N$  lattice.

Let  $v = (a, b)$ ,  $w = (c, d)$ . By applying the division algorithm to the first factor, we can assume that  $c = 0$ . Then, possibly by flipping the sign, we may assume that  $a > 0$  ( $a \neq 0$  as  $v, w$  are linearly independent) and  $d > 0$ . Applying the division algorithm to the second factor, we may assume that  $0 \leq b < d$ . The index of the lattice generated by  $(a, b)$  and  $(0, d)$  is  $ad$ , and its intersection with  $0 \times \mathbb{Z}$  is  $d\mathbb{Z}$ . Therefore, it follows that  $d = N$  and  $a = 1$ , and you are free to choose  $0 \leq b < N$ . Therefore, there are exactly  $N$  many such subgroups.

- (3) Note that a maximal totally ramified abelian extension of  $\mathbb{Q}_3$  is of degree 2 over  $\mathbb{Q}_3$ . Thus, as in Exercise 4.6, we know that there are exactly 3 quadratic extensions of  $\mathbb{Q}_3$ , corresponding to nontrivial elements of  $\mathbb{Q}_3^\times / (\mathbb{Q}_3^\times)^2 = 3^{\mathbb{Z}/2\mathbb{Z}} \times \{1, 2 \pmod{3}\}$ . Therefore, the 3 quadratic extensions are  $\mathbb{Q}_3(\sqrt{2})$ ,  $\mathbb{Q}_3(\sqrt{3})$ ,  $\mathbb{Q}_3(\sqrt{6})$ . As  $X^2 - 2$  is irreducible in  $\mathbb{F}_3[X]$ , it follows that  $\mathbb{Q}_3(\sqrt{2})$  is unramified over  $\mathbb{Q}_3$ , and the other two are ramified. By the degree

<sup>1</sup>You might be confused why there are more than one maximal such extensions, unlike the case of a maximal tamely ramified extension (which is unique). The point is that, even though the tamely-ramified-ness is preserved under compositum, the **totally-ramified-ness** is not preserved under compositum.

reason, it is tamely ramified, and it must be totally ramified. As  $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\frac{-3}{\sqrt{3}})$ , and as  $-3 = 6$  as elements of  $\mathbb{Q}_3^\times/(\mathbb{Q}_3^\times)^2$ , it follows that  $\mathbb{Q}_3(\sqrt{6})$  is  $\mathbb{Q}_3(\zeta_3)$ , and  $\mathbb{Q}_3(\sqrt{3})$  is the other maximal totally ramified abelian extension of  $\mathbb{Q}_3$ . □

**Question 3.** Let  $K/\mathbb{Q}_p$  be a finite extension. For a finite extension  $L/K$  of degree  $n$ , show that

$$f_{L/K} \leq e_{K/\mathbb{Q}_p} \left( v_p(n) + \frac{1}{p-1} \right) + 1.$$

**Hint.** Let  $\pi$  be a uniformizer of  $K$  and  $N = \left\lfloor e_{K/\mathbb{Q}_p} \left( v_p(n) + \frac{1}{p-1} \right) + 1 \right\rfloor$ . Use Exercise 4.4 to deduce that every element in  $1 + \pi^N \mathcal{O}_K$  is an  $n$ -th power. Then, use local class field theory.

*Proof.* Let  $M = \left\lfloor \frac{e_{K/\mathbb{Q}_p}}{p-1} + 1 \right\rfloor$ . By Exercise 4.4,  $1 + \pi^M \mathcal{O}_K \cong \pi^M \mathcal{O}_K$  where the left side group is multiplicative and the right side group is additive. Let  $x \in 1 + \pi^N \mathcal{O}_K$ . Then,

$$\frac{\log(x)}{n} \in \frac{\pi^N}{n} \mathcal{O}_K = \frac{\pi^N}{p^{v_p(n)}} \mathcal{O}_K = \frac{\pi^N}{\pi^{e_{K/\mathbb{Q}_p} v_p(n)}} \mathcal{O}_K = \pi^M \mathcal{O}_K.$$

Therefore,  $\exp(\log(x)/n)^n = x$ , which implies that every element in  $1 + \pi^N \mathcal{O}_K$  is an  $n$ -th power.

Note that  $N_{L/K}(L^\times)$  is of index  $n$  inside  $K^\times$ . Therefore, it must contain  $(K^\times)^n$ , and therefore  $1 + \pi^N \mathcal{O}_K$ . Therefore,  $f_{L/K} \leq N$ , as desired. □